

Disclaimer

- This training resource was developed to be used with the CZ ID platform
- In October, 2026, stewardship of CZ ID transitioned to UCSF and the platform was renamed to Seqtold
- These training materials are not being actively maintained and may not reflect differences between CZ ID and Seqtold, subsequent changes to the platform, or current methods and best practices in metagenomic sequencing and analysis



Generating heatmaps using ONT data from CZ ID

Web-app to create heatmaps

- CZ ID does not have a heatmap visualization feature for the ONT pipeline
- [This web app](#) is designed to mimic this functionality using sample taxon reports downloaded from CZ ID ([Github link](#))

Step 1: Download sample taxon reports

The screenshot shows the iD CZ ID web interface. A modal titled "Select a Download Type" is open, displaying options for downloading data for 36 selected samples. The modal lists four report types: Sample Metadata, Sample Taxon Reports (selected), Combined Sample Taxon Results, and Samples Overview. A "Start Generating Download" button is at the bottom of the modal, with a note that larger files may take hours to generate. The background interface includes a search bar, filters for Taxon and Metadata, and a table of sample data.

Select a Download Type
36 samples selected

REPORTS

- ☐ **Sample Metadata** (sample_metadata.csv)
User-uploaded metadata, including sample collection location, collection date, sample type
- ☒ **Sample Taxon Reports** (.csv)
Computed metrics (e.g. total bases, bPM) and metadata for each taxon identified in the sample
- ☐ **Combined Sample Taxon Results** (.csv)
The value of a particular metric (e.g. total bases, bPM) for all taxa in all selected samples, combined into a single file
- ☐ **Samples Overview** (.csv)
Sample QC metrics (e.g. percent reads passing QC) and other summary statistics

Start Generating Download

Downloads for larger files can take multiple hours to generate.

Background Interface Details:

- Header:** iD CZ ID, My Data, Public, Upload, Gowtham Thakku
- Search:** Search My Data...
- Filters:** Taxon Filters (TAXON, ANNOTATION), Metadata Filters (LOCATION, TIMEFRAME, VISIBILITY, HOST, SAMPLE TYPE)
- Table:** 36 samples selected. Visible rows include RLB2, RLB1, and RLB11_control1_LongA... with columns for sample name, status (COMPLETE), date (2026-06-11), host (Human), location (Bolivia), and metrics (1,978, 77.43%).
- Right Panel:** DESCRIPTION (No description), OVERALL (Samples: 59, Projects: 1, Avg. reads per sample: 12,134,366, Avg. reads passing filters per sample: 1,409,879), DATE CREATED (2026-06-01, 2026-08-07), METADATA (Host: Human, 47).

Step 2: Open the app and upload reports

[Link to web app](#)

Heatmap Parameters

Upload CZID Sample Taxon Reports (CSV) ?

 Upload

200MB per file • CSV

Filter Settings

Metric to display ?

nt_bpm

Taxonomic Level ?

Species ×

Categories ?

viruses ×

CZID Nanopore Heatmap Generator

Upload your CZID sample taxon reports to generate an interactive heatmap. Adjust parameters in the sidebar to customize your visualization.

 [Upload CZID sample taxon report CSV files using the sidebar to get started](#)

Instructions

How to use this app:

1. **Download your data from CZID:**
 - Use the CZ ID bulk download workflow to download Sample Taxon Reports
 - Navigate to https://czid.org/bulk_downloads
 - Download and unzip the CSV reports
2. **Upload your files:**
 - Use the file uploader in the sidebar
 - Select multiple CSV files from your CZID download



Heatmap shows abundance with threshold options on the left panel

Taxonomic Level

Species

Categories

viruses

Threshold Filters

Min NT bpm

10.00

Min NR bpm

0.00

Min reads (r)

0

Min NT Contigs

0



Optionally select samples to include in a background model

The screenshot shows a web interface with a sidebar on the left and a main content area on the right. The sidebar contains three sections: 'Taxonomic Level' with a 'Species' filter, 'Categories' with a 'viruses' filter, and 'Threshold Filters' with three input fields: 'Min NT bPM' (10.00), 'Min NR bPM' (0.00), and 'Min reads (r)' (0). The main content area has a 'Fork' button and a green notification box stating '✓ 10 file(s) uploaded successfully'. Below this is a button 'View uploaded files'. The section 'Background Model (Optional)' is highlighted, with a description 'Select background samples to calculate z-scores for anomaly detection.' and a dropdown menu 'Select Background Samples' set to 'Choose options'. A blue callout box next to the dropdown says 'Select background samples to enable z-score filtering'.

Taxonomic Level

Species

Categories

viruses

Threshold Filters

Min NT bPM

10.00

Min NR bPM

0.00

Min reads (r)

0

Fork

✓ 10 file(s) uploaded successfully

> View uploaded files

Background Model (Optional)

Select background samples to calculate z-scores for anomaly detection.

Select Background Samples

Choose options

Select background samples to enable z-score filtering

The background model uses the same Z-score calculation as the Standard model used in the CZ ID Illumina pipeline, except using bPM as the abundance metric instead of rPM. See DL7-O_Background models for ONT slides for more information

Set Z-score threshold for use in the background model

The screenshot shows a web interface with a sidebar on the left and a main content area on the right. The sidebar contains three sections: 'Taxonomic Level' with a 'Species' tag, 'Categories' with a 'viruses' tag, and 'Threshold Filters' with three input fields: 'Min NT bPM' (10.00), 'Min NR bPM' (0.00), and 'Min reads (r)' (0). The main content area has a 'View uploaded files' button at the top. Below it, the 'Background Model (Optional)' section is titled, followed by the instruction 'Select background samples to calculate z-scores for anomaly detection.' There are two input fields: 'Select Background Samples' and 'Min Z-score Threshold'. The 'Select Background Samples' field contains four tags: 'RLB02_Water_NP_filtered', 'RLB05_Water_NP_filtered', 'RLB05_Water_Serum_filtered', and 'RLB02_Water_Serum_filtered'. The 'Min Z-score Threshold' field is a slider set to 1.00.

Taxonomic Level ?

Species × ▼

Categories ?

viruses × ▼

Threshold Filters

Min NT bPM ?

10.00 − +

Min NR bPM ?

0.00 − +

Min reads (r) ?

0 − +

Fork 🔄 ⋮

> View uploaded files

Background Model (Optional)

Select background samples to calculate z-scores for anomaly detection.

Select Background Samples ? Min Z-score Threshold ?

1.00 − +

RLB02_Water_NP_filtered ×

RLB05_Water_NP_filtered ×

RLB05_Water_Serum_filtered ×

RLB02_Water_Serum_filtered ×

Choice of Z-score threshold depends on the levels of contamination observed, but a good starting point is a threshold of 1.00

Heatmap can be download as .png image file or .csv data file

nc_001101.1

Taxonomic Level

Species ×

Categories

viruses ×

Threshold Filters

Min NT bPM

10.00 - +

Min NR bPM

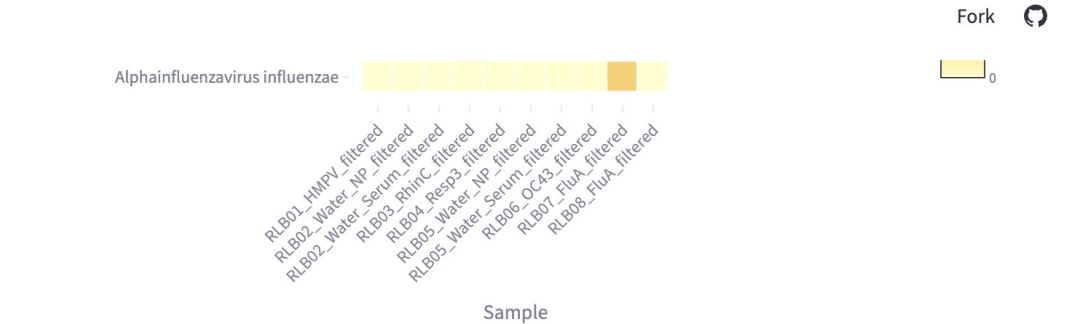
0.00 - +

Min reads (r)

0 - +

Min NT Contigs

0 - +



Download Options

Download as PNG

Download Data as CSV

Click the camera icon in the plotly toolbar above to download as PNG

> View Summary Statistics

Now you can try it!

[Web app](#)
[Github link](#)